Unknown sides

1 Find the number that completes a Pythagorean triple given the following:

a The two largest numbers are 10 and 8.

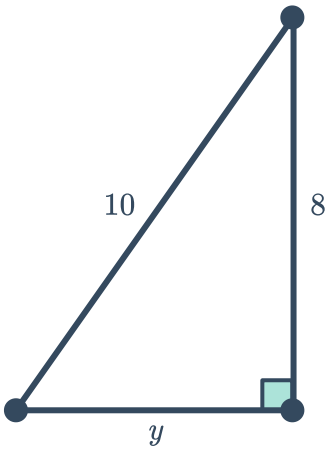
b The largest and smallest numbers are 41 and 9.

2 For the following right-angled triangles:

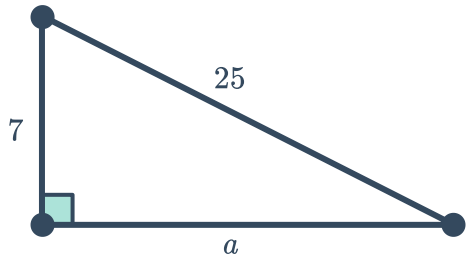
i Write the equation that the sides of the triangle satisfy.

ii Solve the equation to find the length of the unknown side.

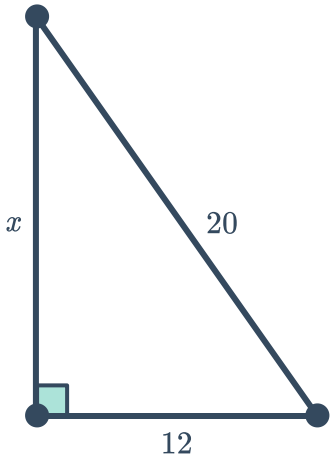
a



b



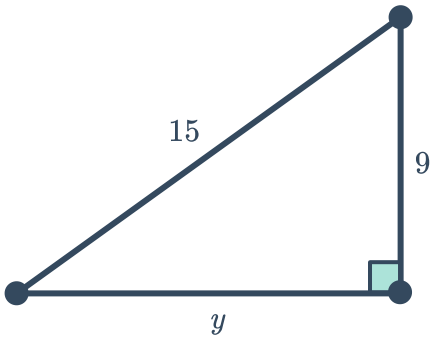
c



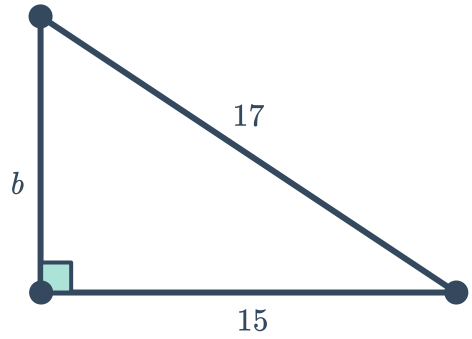
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3 Find the length of the unknown side of the following triangles:

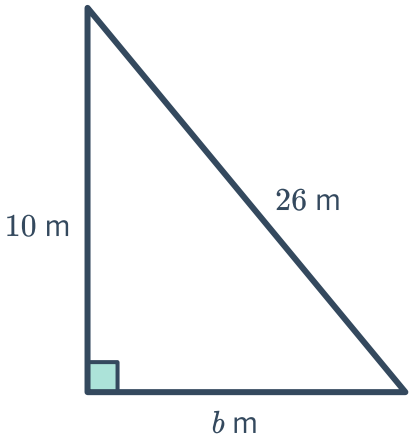
a



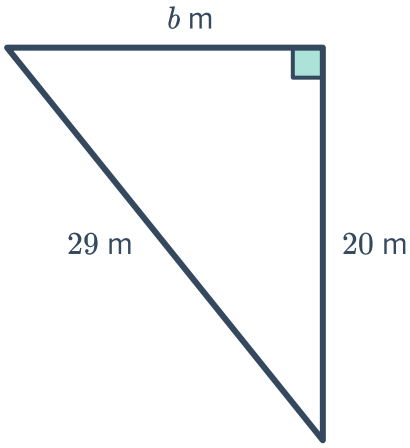
b



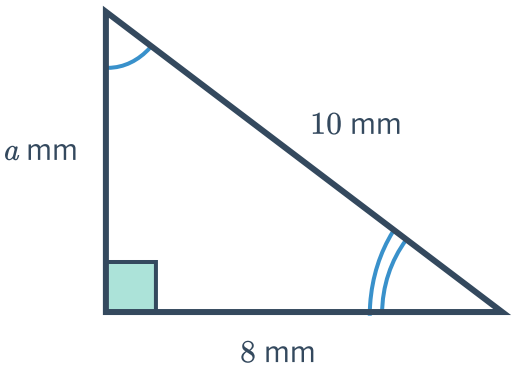
c



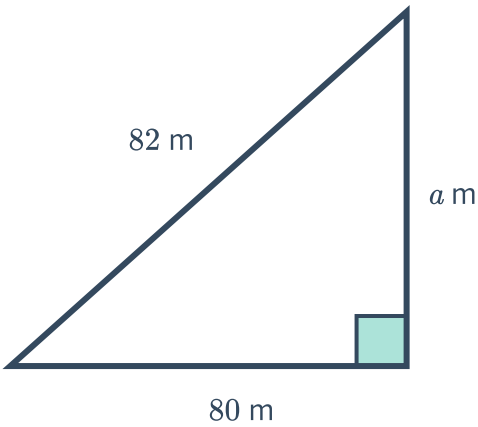
d



e



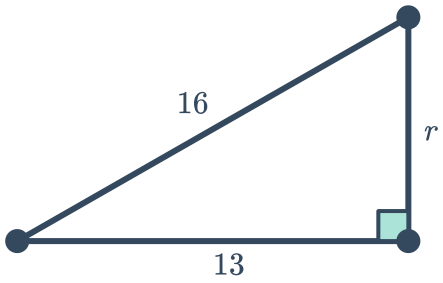
f



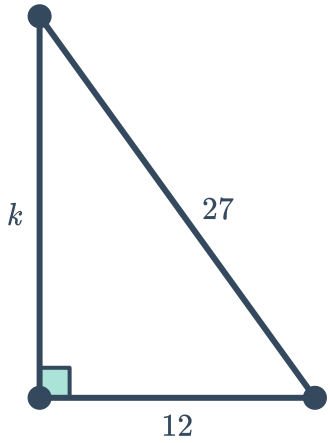
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4 Find the length of the unknown side of the following triangles, correct to two decimal places:

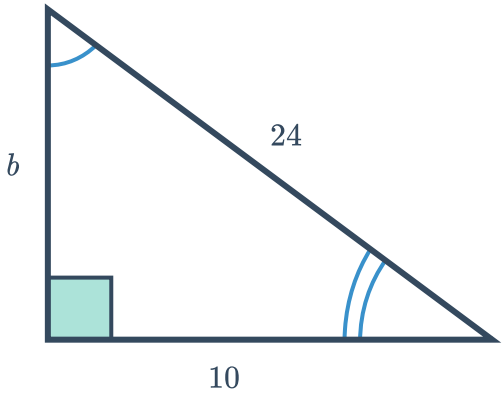
a



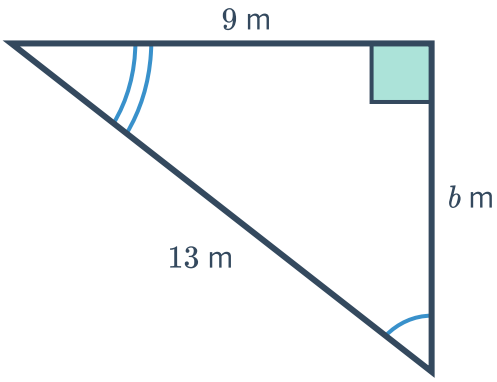
b



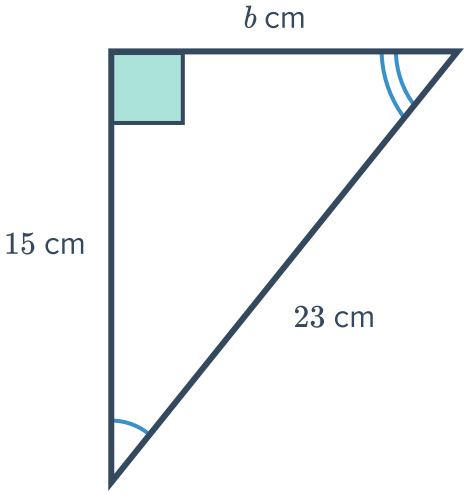
c



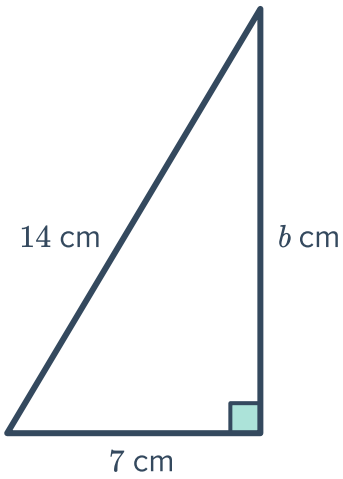
d



e

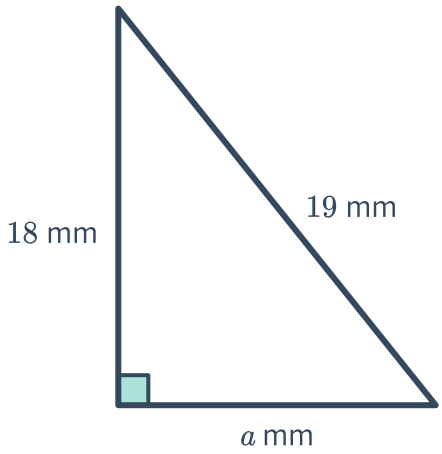


f

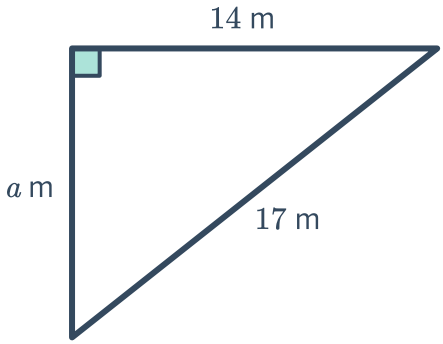


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g

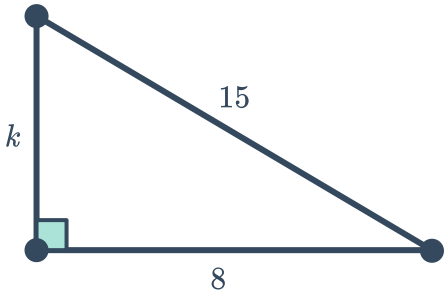


h

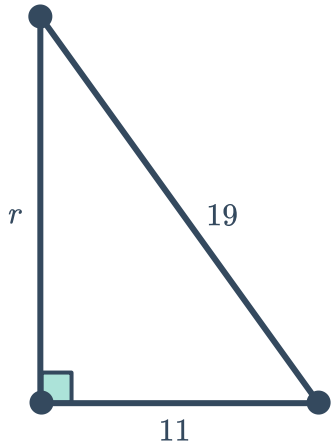


5 Find the length of the unknown side of the following triangles, giving your answer as a surd:

a



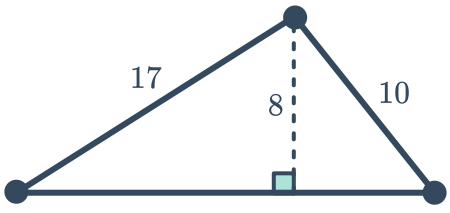
b



6 The two short sides of a triangle have lengths of 10 and 17. Sarah notices that she can divide the triangle into two smaller right-angled triangles, each with a height of 8.

a Find the base lengths of the two right-angled triangles.

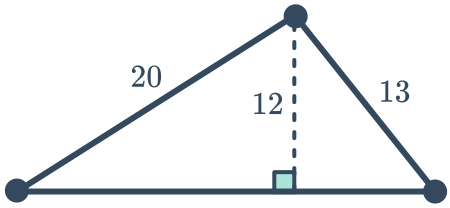
b Find the base length of the original triangle.



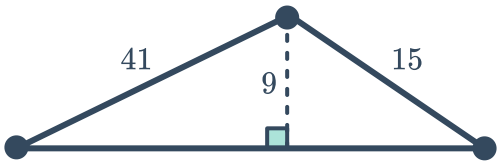
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7 Find the base length of the following triangles:

a



b



Perimeter

8 Consider a right-angled triangle with hypotenuse of length $15\,\text{cm}$ and another side of length $4\,\text{cm}$.

a Find the length of the other side, correct to two decimal places.

b Find the perimeter, correct to two decimal places.

9 Consider a rectangle with a diagonal of length $25\,\text{cm}$ and a side of length $24\,\text{cm}$.

a Find the length of the other side.

b Find the perimeter.

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